Ezenia Diaz-Lembert

(774) 303-3666 | ezenia.diazlembert@gmail.com | linkedin.com/in/ezenia-diaz-lembert/

EDUCATION

Harvard University | Cambridge, MA *Sep 2020-Feb 2025*
Bachelor of Science in Electrical Engineering

Relevant Coursework: Design of VLSI Circuits and Systems; Advanced Mixed-Signal CMOS Circuits; Circuits, Devices, and Transduction; Systems and Control; Applied Electromagnetism; Neurophysiology and Neural Interfaces

TECHNICAL SKILLS

Programs and languages: SystemVerilog, Vivado, Cadence Virtuoso/Layout, LTspice, MATLAB, Python, C/C++, Bash, GIS

Lab and validation: photolithography, oscilloscope, waveform generator, hardware-in-the-loop testing

PROJECTS

The Current Collective *Jan 2026-Present*

- Created and host a podcast explaining emerging research in tech and consumer electronics to a general audience.

Dynamic Equalization Algorithm *Jan 2024-Dec 2024*

- Designed a real-time adaptive filter for personalized music equalization as a senior capstone project.
- Implemented a Convolutional Neural Network and spectral feature extraction on input audio signals for dynamic equalization, achieving 83.2% accuracy.

MIPS Multicycle Processor *Jan 2023-May 2023*

- Designed and implemented a datapath and control unit for a multicycle processor in SystemVerilog.

WORK EXPERIENCE

AVANGRID | Augusta, ME *July 2025-Aug 2026*

Energy Control System Designer - Power Systems Applications Analyst

- Designed and automated a transmission network model for Central Maine Power's energy management system.
- Defined technical requirements, strategies, and procedures for transmission and distribution systems.
- Wrote a Python script to synchronize data across the energy management system and distribution one-line diagrams, reducing reconciliation time by 15 hours/month.

BOSE Corporation | Framingham, MA *Jan 2025-May 2025*
Global R&D Consumer Systems and Architecture Co-op

- Built biquad filter models that emulate headset fits and synthesized them for FPGA-in-the-loop testing.
- Designed and validated test stimuli across frequencies to characterize filter response.

Eaton Corporation | Moon Township, PA *Jun 2024-Aug 2024*
Electrical Engineering Intern

- Improved Total Harmonic Distortion computation accuracy from 7th to 39th current harmonics for the PCD-Air Circuit Breaker team by developing C code in IAR Tool.
- Tested real-time THD accuracy using injected test signals from a wave generator and oscilloscope FFT projections.

BAE Systems | Merrimack, NH *Sep 2023-Jan 2024*
Electrical Engineering Intern

- Conducted analog circuit design and analysis, simulations, and layout for the Mixed Signal Integrated Circuit team.
- Ran Monte Carlo simulations for phase-locked loop systems.

United States DOE Office of Technology Transitions Entrepreneurship Program | Hybrid *May 2022-Aug 2022*
Summer Intern

- Led customer discovery interviews, researched a $3.26 billion market, and built a business model canvas to produce a 4-year commercialization plan for a Sandia National Laboratory technology.

LEADERSHIP

Harvard SEAS Society of Hispanic Professional Engineers | Cambridge, MA *Sep 2020-Dec 2023*
Corporate Liaison Officer

- Managed communications with corporate partners, applied for grants, and organized STEM events on campus.